COMPUTER NETWORK (BCSC0008)

ASSIGNMENT -1

Q1. A university campus has two labs:

- Lab A is wired with Ethernet using CSMA/CD.
- Lab B uses Wi-Fi (802.11) with CSMA/CA enabled.

During peak hours, both labs have 20 students streaming videos simultaneously. Which lab is expected to have higher throughput efficiency and why?

Q2. A system uses Pure ALOHA with frame time = 100 ms and channel rate = 200 kbps.

- (a) What is the maximum channel utilization?
- (b) What is the maximum throughput in kbps?

Q3. A sender uses Stop-and-Wait flow control to send 1000-byte frames over a 1 Mbps link. Propagation delay = 20 ms one way.

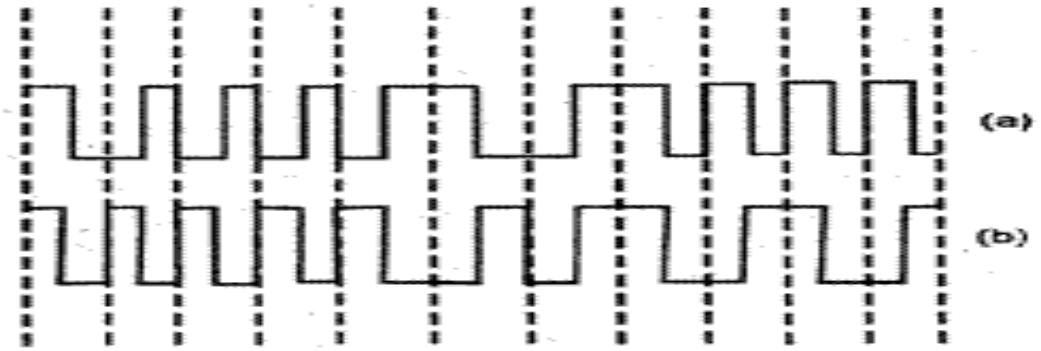
- Calculate the link utilization.
- Suggest how efficiency can be improved

Q4. Station A sends frames to Station B at 1 Mbps, but Station B can only process data at 500 kbps**.**

- What problem will occur if no flow control is used?
- Which mechanism prevents this?

Q5. Selective Repeat requires $N \leq 2^m$ sequence numbers and window size $W \leq 2^{m-1}$. For a protocol with m=4 bits for sequence numbers, what is the maximum window size W for Selective Repeat? If sender wants W = 12, is it allowed?

Q6. In the waveform (a) given below, a bit stream is encoded by Manchester encoding scheme. The same bit stream is encoded in a different coding scheme in wave form (b). The bit stream and the coding scheme are-



Q7. An error correcting code has the following code words: 00000000, 00001111, 01010101, 10101010, 11110000. What is the maximum number of bit errors that can be corrected?

Q8. The minimum frame size required for a CSMA/CD based computer network running at 1 Gbps on a 200m cable with a link speed of 2×10^8m/s is-

Q9- A bit stream 1101011011 need to be transmitted using the standard CRC method. The generator polynomial is x^4+x+1. What is the actual bit string transmitted?

Q10- Data bits (4) = 1011. Encode using **Hamming (7,4)** (parity bits at positions 1,2,4). Give the 7-bit codeword.